Assignment 2

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Please note that

- **Released date**: 2023.6.16, 19.00 p.m.
- **Due date**: 2023.6.23, 11.59 p.m.
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like `"XXXXXXXXX_HW2.pdf"` (Your student ID + HW No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size $\leq$ 10MB).
- Please make sure that your submitted file is clear and readable. Submitted file that can not be opened or not readable will get 0 point.

Question 1. Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$, calculate $(\mathbf{AB})^{10}$.

Question 2. Let $\mathbf{a}_1 = [1,1]^T, \mathbf{a}_2 = [1,-1]^T$, and matrix $\mathbf{A} == \begin{bmatrix} -2 & 6 \\ 0 & 4 \end{bmatrix}$. If $\mathbf{A} = k_1\mathbf{a}_1\mathbf{a}_1^T + k_2\mathbf{a}_1\mathbf{a}_2^T + k_3\mathbf{a}_2\mathbf{a}_1^T + k_4\mathbf{a}_2\mathbf{a}_2^T$, what are the values of $k_i, i = 1, \ldots, 4$.

Question 3. Consider a 4×5 matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5]$. The reduced row-echelon form of $\mathbf{A}$ is

$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{1}$$

Moreover, $\mathbf{x}_0 = (3,2,0,2,0)^T$ is a solution to $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{b} = (0,5,3,4)^T$.

(a) Find the solution set of $\mathbf{Ax} = \mathbf{b}$.

(b) Recover $\mathbf{A}$ if we already know that $\mathbf{a}_1 = (2,1,-3,-2)^T$ and $\mathbf{a}_2 = (-1,2,3,1)^T$.

Question 4. Compute the matrix-matrix multiplication

(a) $diag(1,-2,4)\begin{bmatrix} 3 & 4 & 1 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 4 & 1 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix} diag(1,-2,4).$

Question 5. Let $\mathbf{A} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$.Please compute $\mathbf{A}^2, \mathbf{A}^3$. What will $\mathbf{A}^n$ be?

Question 6. Given $\mathbf{A} \in \mathbb{R}^{2\times 2}$. If $\mathbf{A}$ commutes with $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, show that if $\mathbf{A} = \begin{bmatrix} a & 0 \\ c & a \end{bmatrix}$ for some a and c.

Question 7. Compute the following by using block multiplication (all blocks are $k \times k$).

1. $\begin{bmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}\begin{bmatrix} \mathbf{I} & \mathbf{-X} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$
2. $[\mathbf{I} \quad \mathbf{X}^T][\mathbf{-X} \quad \mathbf{I}]^T$

Question 8. Compute $\mathbf{AB}$ by using the following block partitioning.

$$\mathbf{A} = \left[\begin{array}{cc|cc} 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \quad \mathbf{B} = \left[\begin{array}{cc|c} 1 & -2 & 0 \\ -1 & 0 & 0 \\ \hline 0 & 5 & 2 \\ -1 & 1 & 0 \end{array}\right]$$

Question 9. For any $\mathbf{A} \in \mathbb{R}^{n\times n}, \mathbf{B} \in \mathbb{R}^{n\times n}$, show that $\mathbf{AB} = \mathbf{BA}$ if and only if $(\mathbf{A}+\mathbf{B})(\mathbf{A}-\mathbf{B}) = (\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B})$.

Question 10. Find $\mathbf{A}$ if $(\mathbf{A}^T - 2\mathbf{I})^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

Question 11. (a) Let $\mathbf{A}$ be a nonsingular matrix. Show that $\mathbf{A}^{-1}$ is also nonsingular and $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.

(b) Let $\mathbf{A}$ be a nonsingular matrix. Show that $\mathbf{A}^T$ is also nonsingular and $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$.

Question 12. Given $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}$, find a matrix $\mathbf{B}$ such that $\mathbf{AB} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

Question 13. Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}.$$

(a) Write the elementary matrices $\mathbf{E}_1$, $\mathbf{E}_2$ and $\mathbf{E}_3$ from the given $\mathbf{A}$ under the condition that $\mathbf{E}_1$, $\mathbf{E}_2$ and $\mathbf{E}_3$ contain $-a, -b$ and $-c$ respectively.

(b) Multiply $\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1$ to find the single matrix $\mathbf{E}$ that produces $\mathbf{EA} = \mathbf{I}$.

(c) Multiply $\mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\mathbf{E}_3^{-1}$ to bring back $\mathbf{A}$. You will see that the multipliers a, b, c are mixed up in $\mathbf{E}$ but perfect in $\mathbf{A}$.

Question 14. Find an **LU**-factorization for

$$\begin{bmatrix} -1 & -3 & 1 \\ 1 & 4 & 1 \\ 0 & -2 & -4 \end{bmatrix}.$$

Question 15. Find a permutation matrix $\mathbf{P}$ and an **LU** decomposition of **PA** if $\mathbf{A}$ is

$$\begin{bmatrix} 0 & -2 & 2 \\ 1 & 0 & 4 \\ -1 & 2 & 1 \end{bmatrix}.$$

Question 16. Solve the system $\boldsymbol{Ax} = \boldsymbol{b}$ by the provided LU decomposition of $\boldsymbol{A}$.

$$\boldsymbol{A} = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$